OLD DOMINION
UNIVERSITY

EET 405
Data Communications and Computer Networks

Internet of Things Hands-on Activities with Arduino

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Department of Engineering Technology

Today! Internet of Things (IoT) Hands-on Activities

- Introduction
- Download Folders
  - (1) Github Repository (Recommended)

    https://github.com/muratkuzlu/EET405_S2022_IoT_Notes
  - (2) Google Drive:

    https://drive.google.com/drive/folders/0B2WhaJ0_Fq-5fi01TWJVRzdZeW9VeW5STUJXQzJNbUxOcDlqQl9tUzNDNDFCWVRSNmlwdlE?resourcekey=0-YgSIcYySE9kjPWY2USWlDg
- WiFi Information (SSID and Password)
- Presentation (Summary)
- Hands-on Activities
- Q&A

Outline

- Internet
- Internet of Things (IoT)
- Arduino
- ThingSpeak
- Hands-on Activities



https://medium.com/@otavioguastamacchia/creating-a-simple-iot-case-8102f22908a7

What is the Internet?



https://www.connexusuk.com/high-speed-internet/

- The Internet is a global **web of computers connected to each by communication lines (wired or wireless).**



http://mediatechnologyeducation.pbworks.com/w/page/20693030/The%20Information%20Superhighway

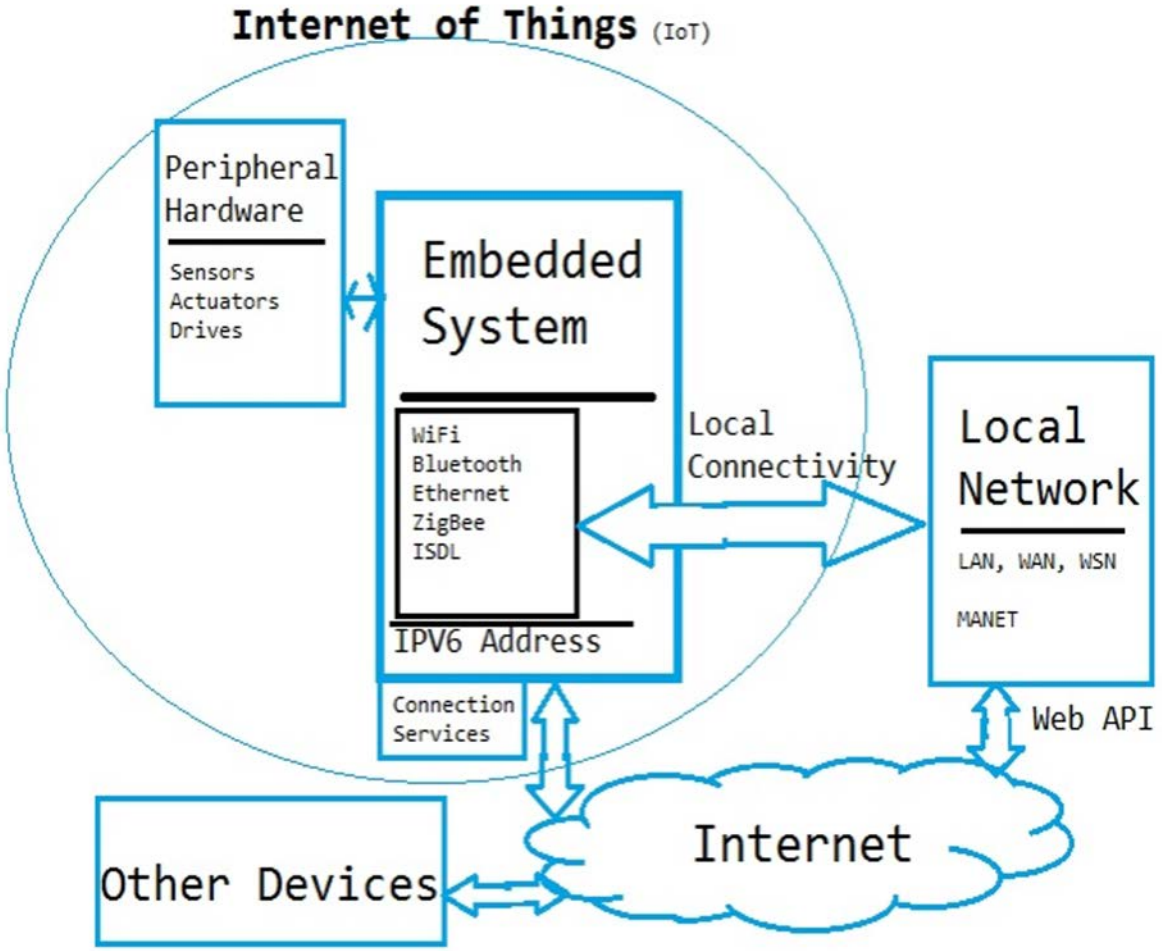
- If you look at a map of big cities, smaller towns, and scattered houses, each is **connected together with roads, railways, etc**. This is similar to the Internet, except with the Internet, wires connect computers.

  The Internet is a superhighway.

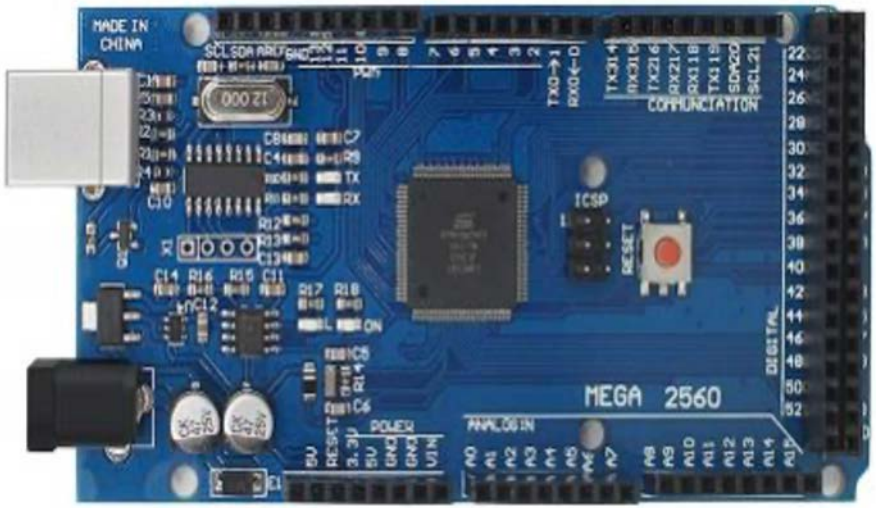
The Internet of Things (IoT)

IoT: You can access anything (devices) from anywhere (through the INTERNET)!

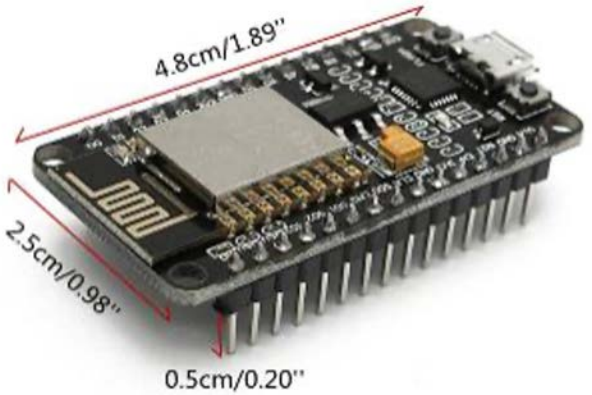
- *Network of physical objects*
- *Embedded System*
- *Network connectivity*
- *Collect and exchange data*



Arduino

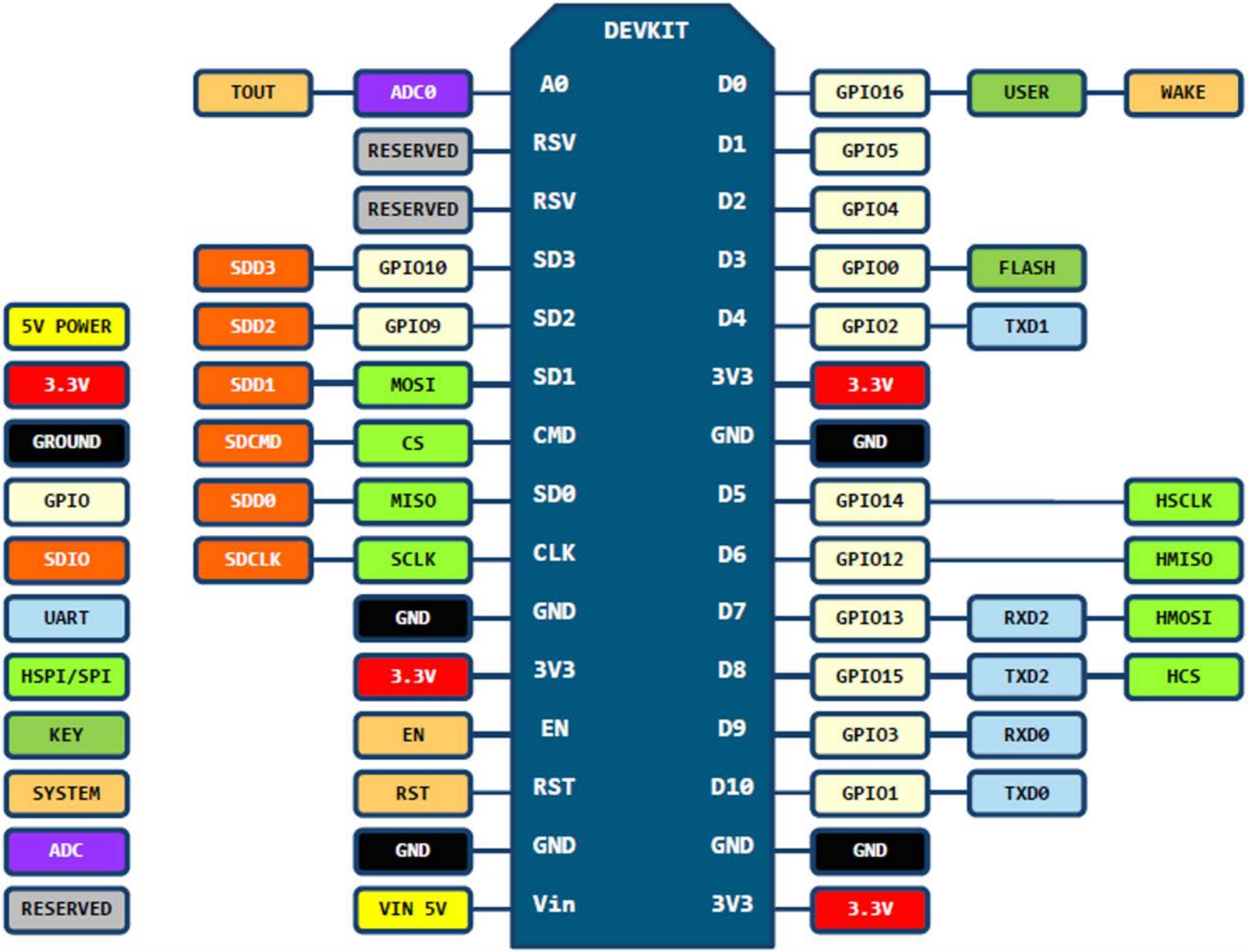


Mega2560 R3 ATmega2560-16AU CH340 Development Board



ESP8266 **NodeMCU** LUA CP2102 ESP-12E

NodeMCU ESP8266 ESP-12E Pinout

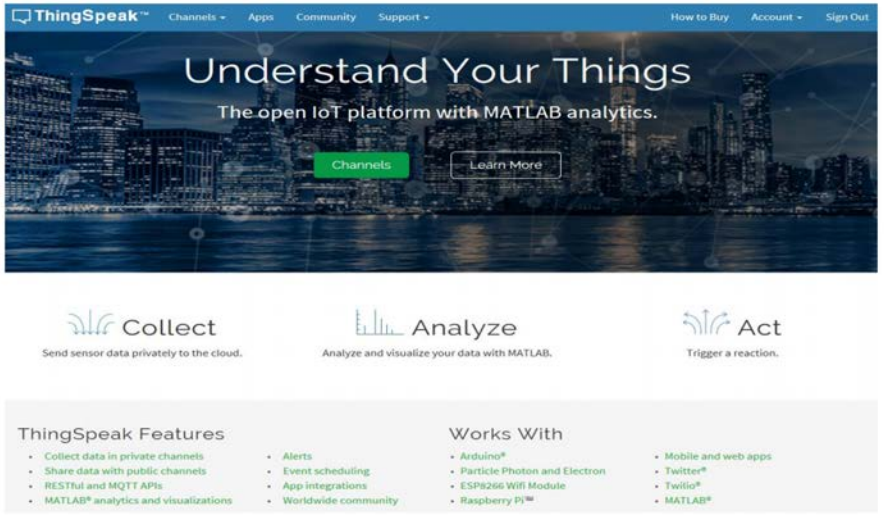


What is ThingSpeak?

- **What is ThingSpeak:**
  ThingSpeak is an IoT analytics platform service that lets you collect and store sensor data in the cloud and develop Internet of Things applications.

- https://thingspeak.com

- **ThingSpeak Channel:**
  'ThingSpeak Channel' is the core element of the ThingSpeak platform. This channel is used to store the real-time data, or the data transferred through various sensors and embedded systems. Data stored at the channel is further used for analysis and visualization.

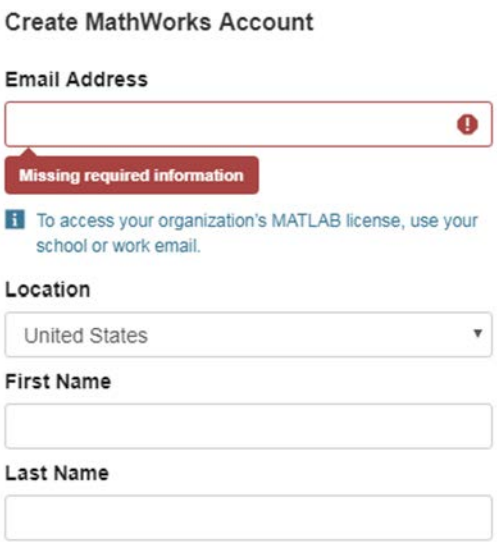


- Software Requirement: Internet
- Hardware Requirement: Arduino.

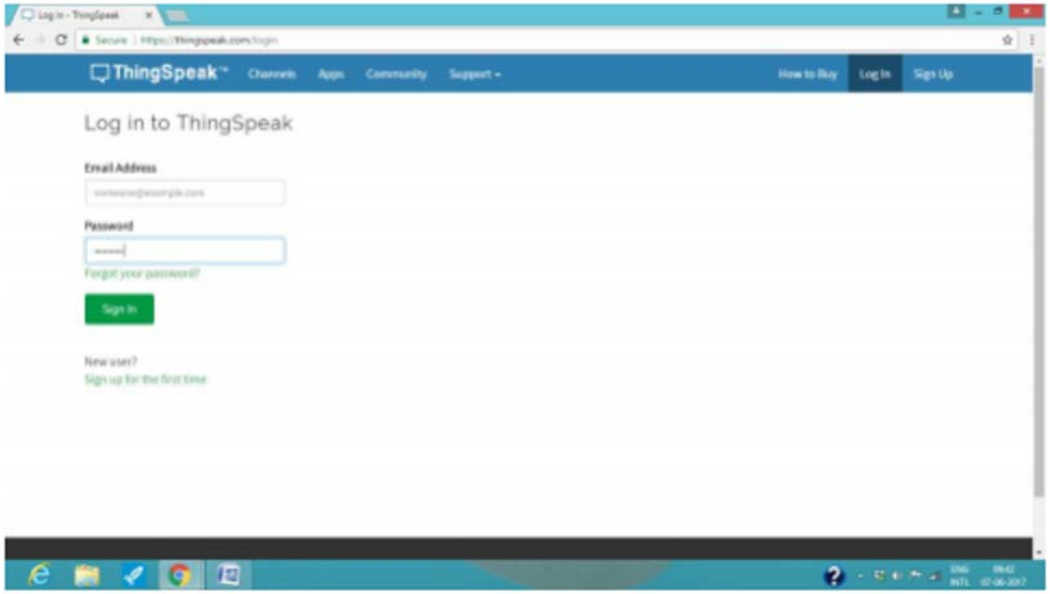
Creating A Channel

- **https://thingspeak.com**

- Before creating a channel, you need to sign into things speak. You can easily sign in either using your either ThingSpeak account or mathswork account, or create a new mathswork account via following link:
- https://thingspeak.com/users/sign_up

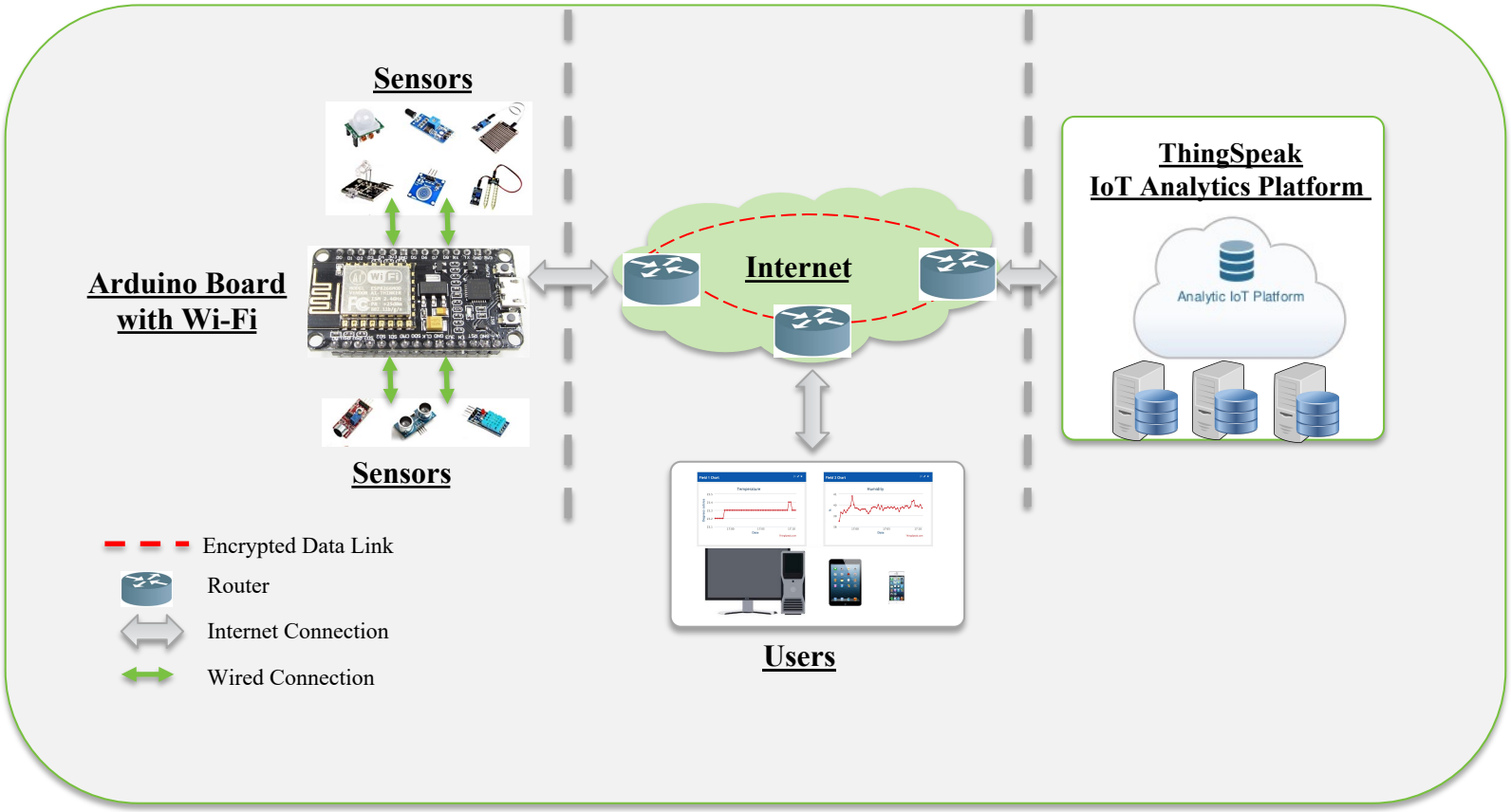


- Login Page



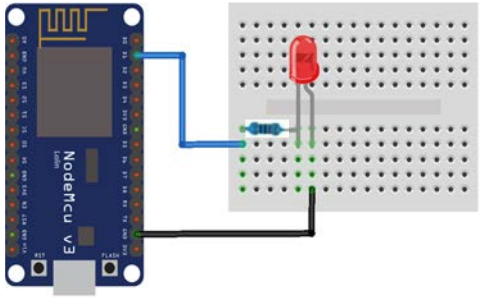
- Email: mkuzlu@hotmail.com
- Password: ODU_Blast2000

Architecture of an IoT-based System



Hands-on Activities

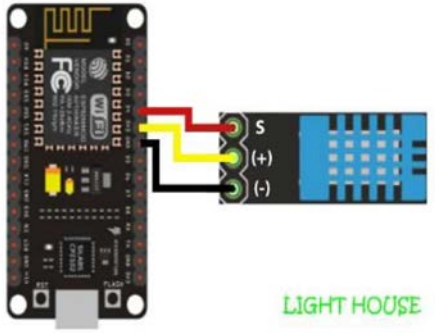


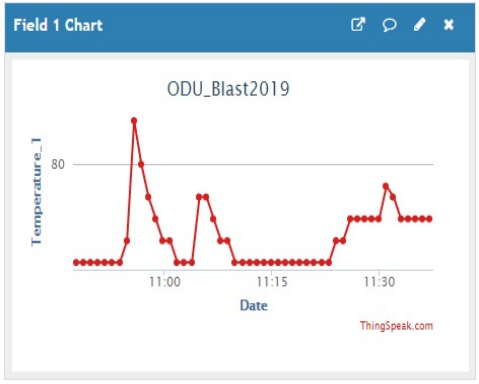


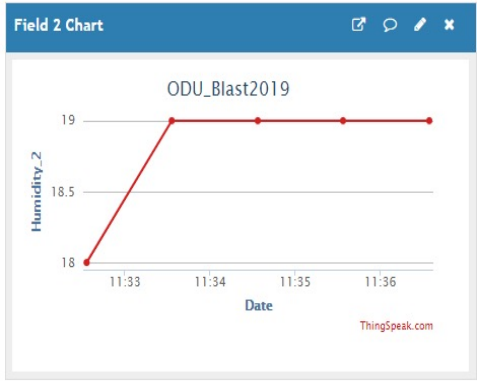

```
#define LED D1 // Led in NodeMCU at pin GPIO5 (D1).

void setup()
{
pinMode(LED, OUTPUT); // set the digital pin as output.
}

void loop()
{
digitalWrite(LED, HIGH); // turn the LED off.
delay(1000);  // wait for 1 second.
digitalWrite(LED, LOW); // turn the LED on.
delay(1000); // wait for 1 second.
}
```





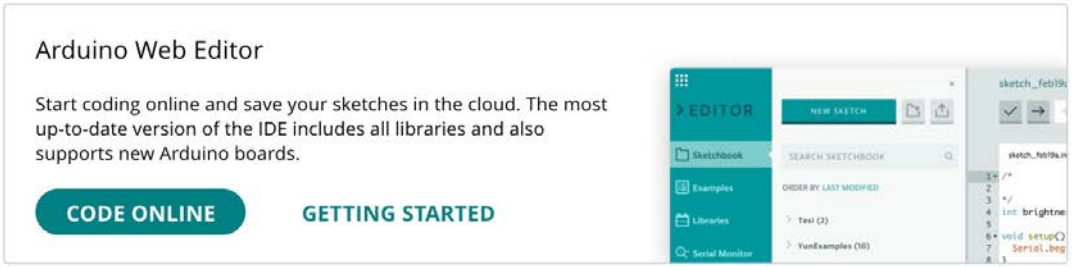


Preparation!

Step 1 - Installing IDE

Installing IDE

- The Arduino Integrated Development Environment (IDE) is the software side of the Arduino platform.
- **Go to https://www.arduino.cc/en/software**





Downloads

Arduino IDE 1.8.15

The open-source Arduino Software (IDE) makes it easy to write code and upload it to the board. This software can be used with any Arduino board.

Refer to the **Getting Started** page for Installation instructions.

SOURCE CODE

Active development of the Arduino software is **hosted by GitHub**. See the instructions for **building the code**. Latest release source code archives are available **here**. The archives are PGP-signed so they can be verified using **this** gpg key.

DOWNLOAD OPTIONS

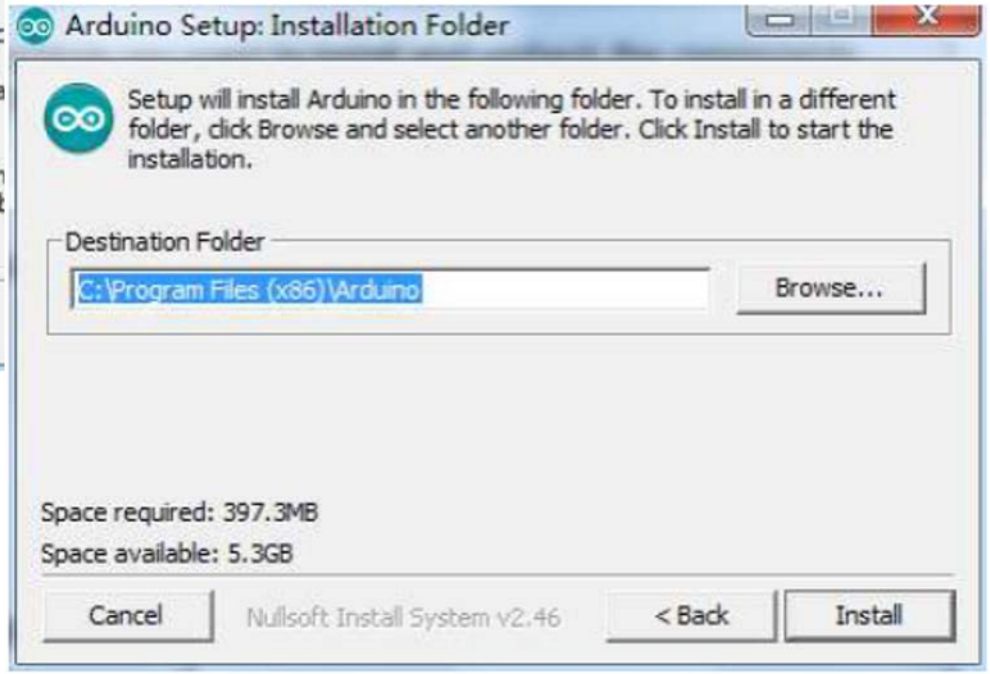
Windows Win 7 and newer
Windows ZIP file
Windows app Win 8.1 or 10 Get
Linux 32 bits
Linux 64 bits
Linux ARM 32 bits
Linux ARM 64 bits
Mac OS X 10.10 or newer

Release Notes Checksums (sha512)

Installing Arduino (Windows)



Click Agree and Install...

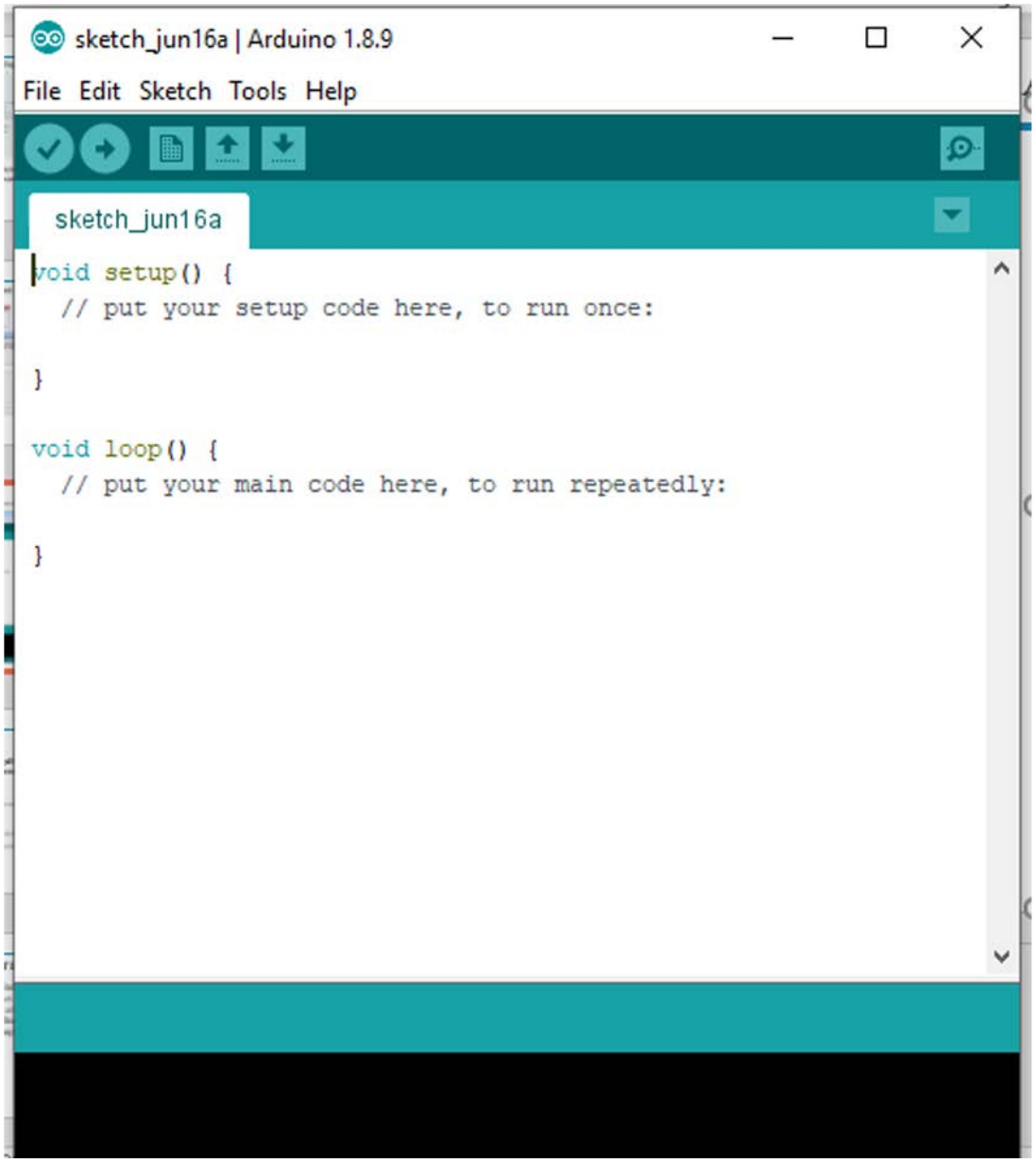


The following Icon appears on the desktop



Arduino IDE

Double-click to enter the desired development environment



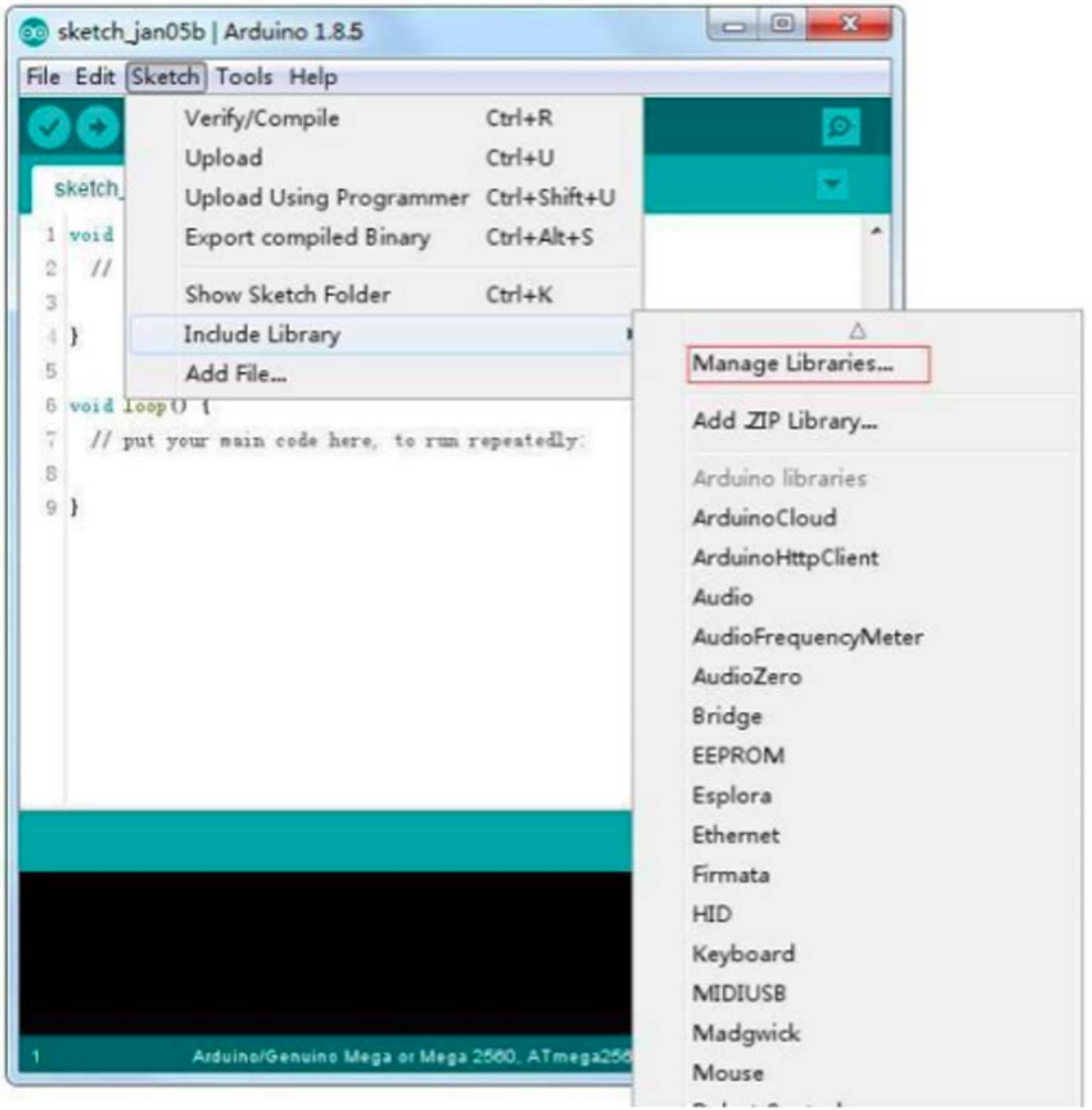
Step 2 - Add Libraries

Add Libraries

How to Install a Library?

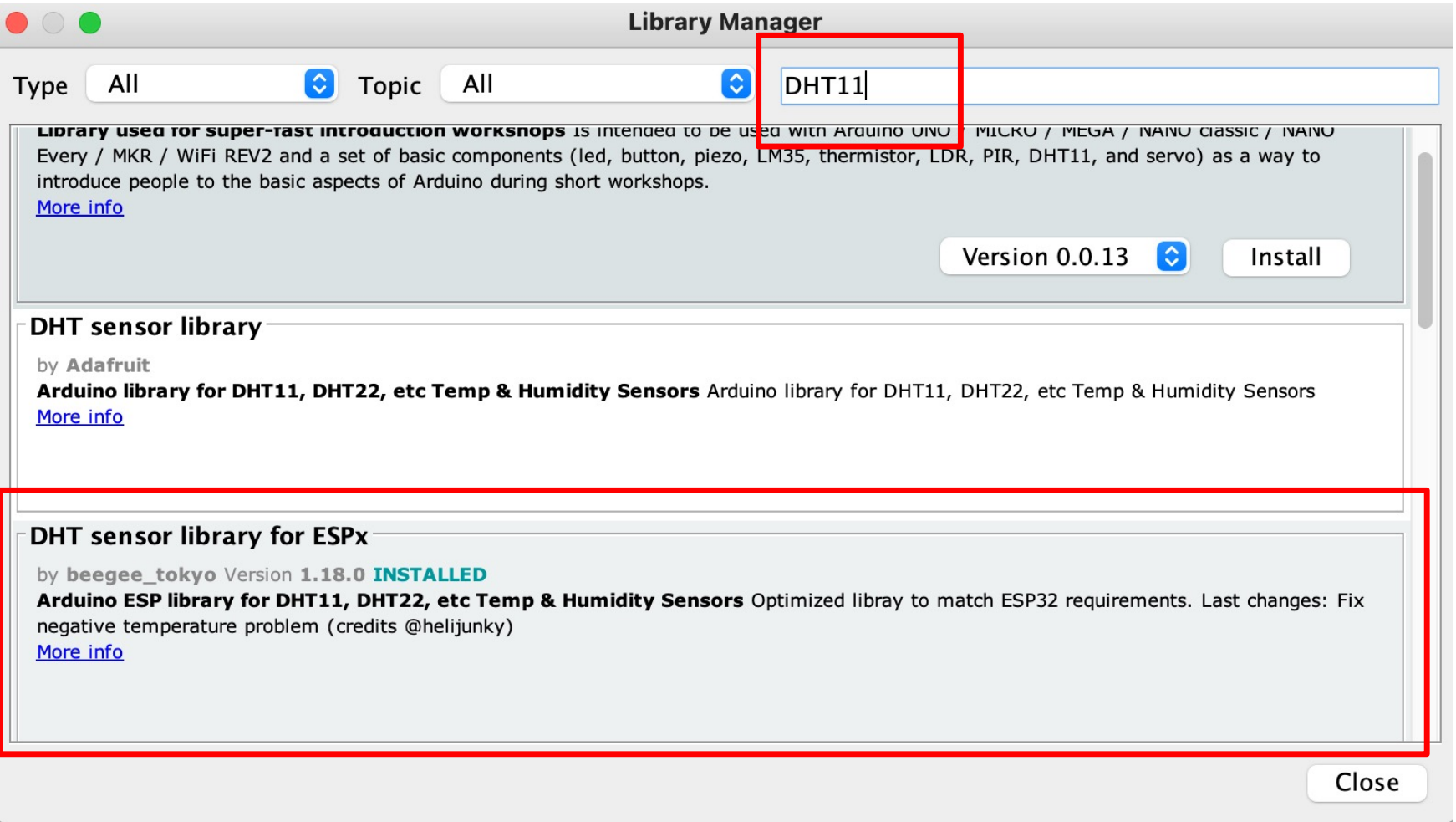
Using the Library Manager
To install a new library into your Arduino IDE you can use the Library Manager (available from IDE version 1.8.0).

Open the IDE and click to the "Sketch" menu and then Include Library > Manage Libraries.



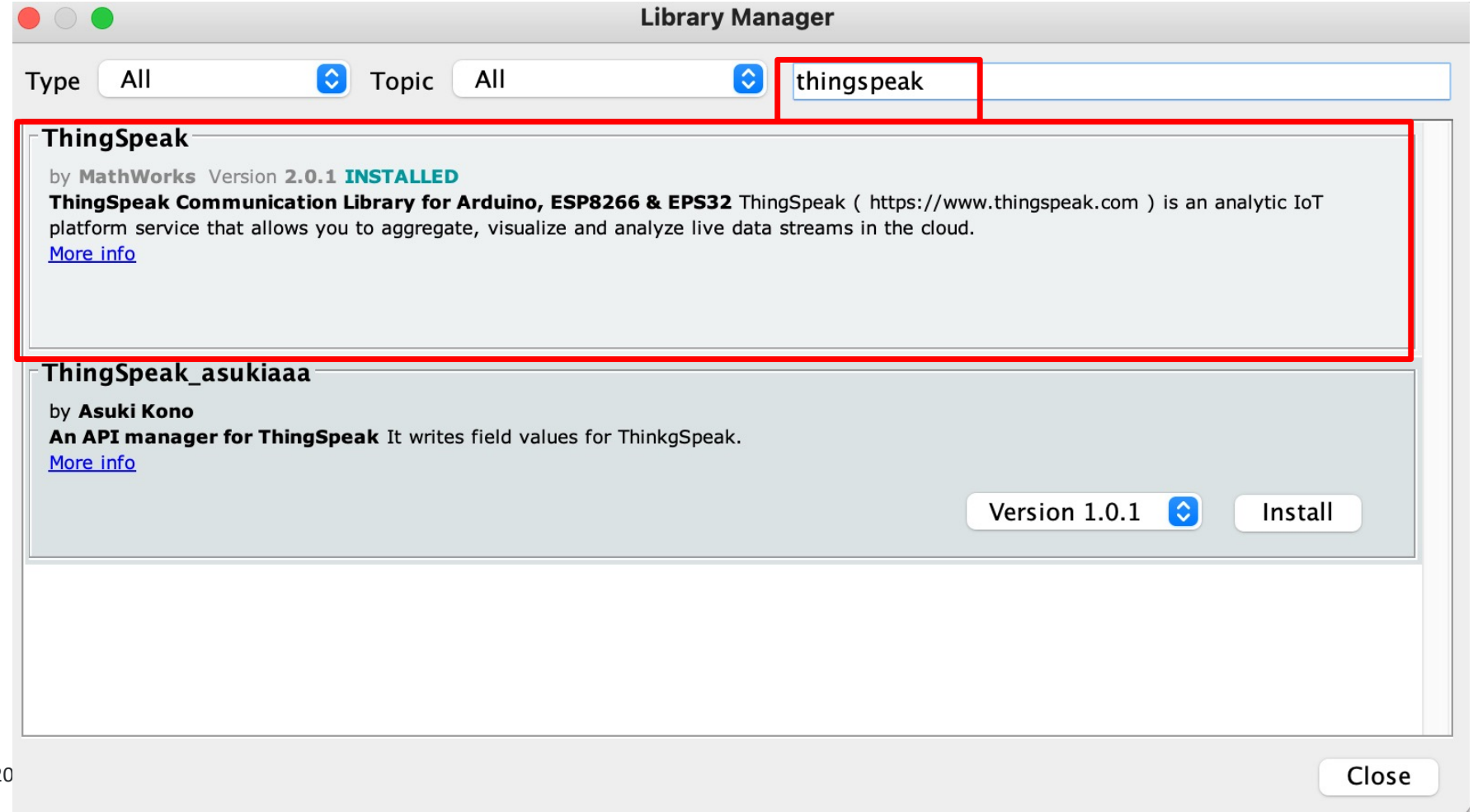
Add Libraries- DHT11

Open the IDE and click to the "Sketch" menu and then Include Library > Manage Libraries > Type “DHT11” > Install “DHT sensor library” and “DHT sensor library for ESPx”



Add Libraries- Arduino

- The Arduino IDE needs to have the ThingSpeak library installed in order for your devices to know how to send data to ThingSpeak. In the Arduino IDE, choose Sketch, Include Library, and Manage Libraries. Search for “thingspeak” and click Install.



Step 3 - NodeMCU Setup

NodeMCU Setup

- *Additional Board Manager URL:*
  - ***http://arduino.esp8266.com/stable/package_esp8266com_index.json***

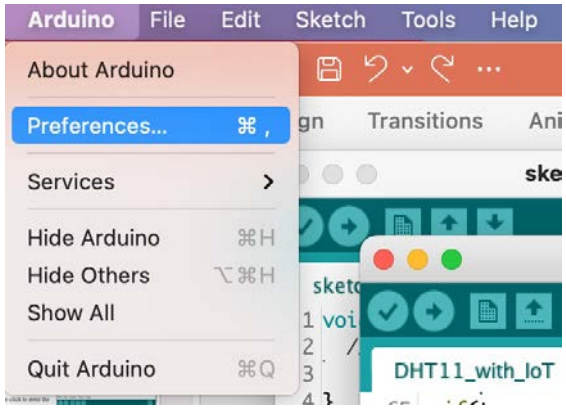
Installing with Boards Manager

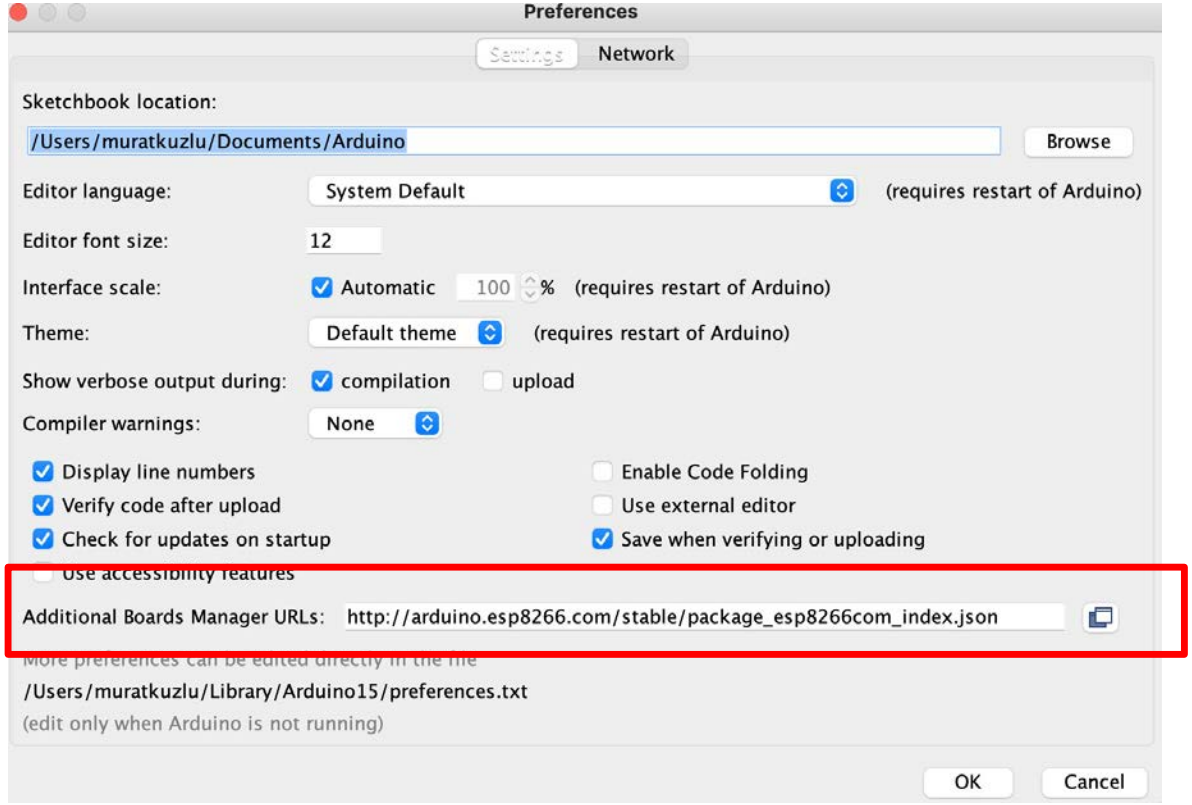
Starting with 1.6.4, Arduino allows installation of third-party platform packages using Boards Manager. We have packages available for Windows, Mac OS, and Linux (32 and 64 bit).

- Install Arduino 1.6.8 from the Arduino website.
- Start Arduino and open Preferences window.
- Enter `http://arduino.esp8266.com/stable/package_esp8266com_index.json` into *Additional Board Manager URLs* field. You can add multiple URLs, separating them with commas.
- Open Boards Manager from Tools > Board menu and install *esp8266* platform (and don't forget to select your ESP8266 board from Tools > Board menu after installation).

NodeMCU Setup

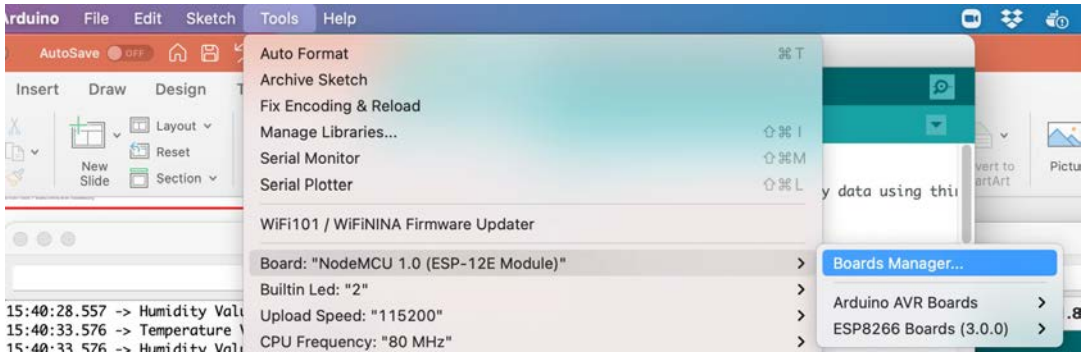
- Add the link to the *Additional Board Manager URLs*

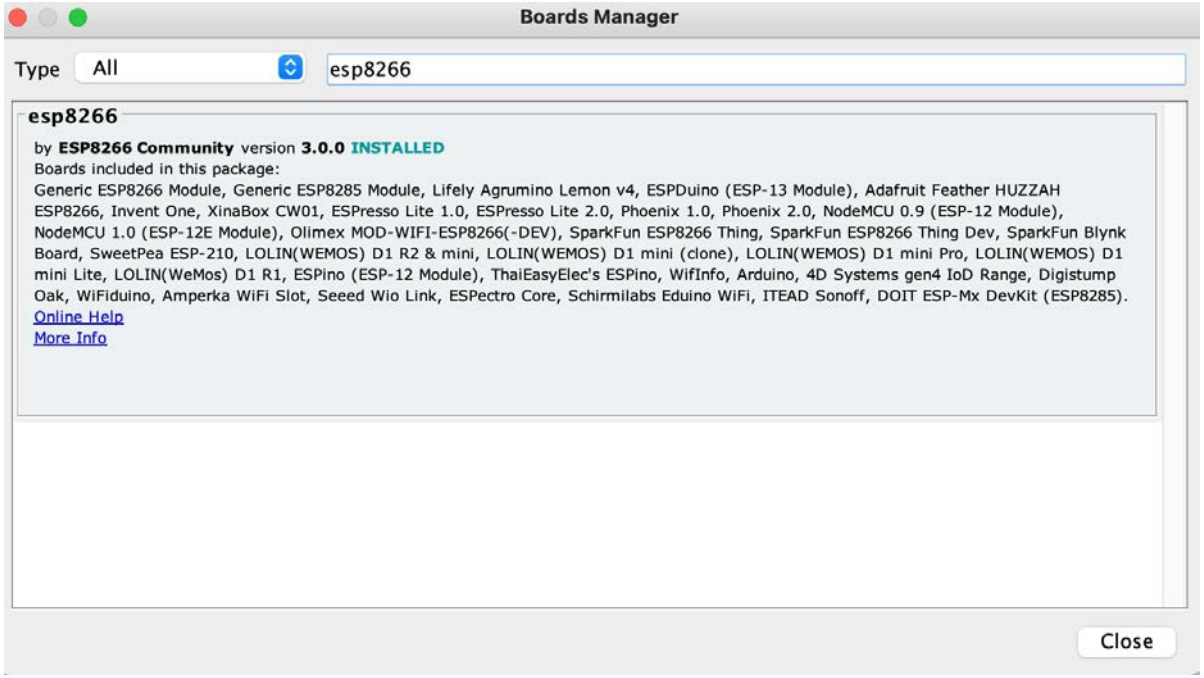




NodeMCU Setup

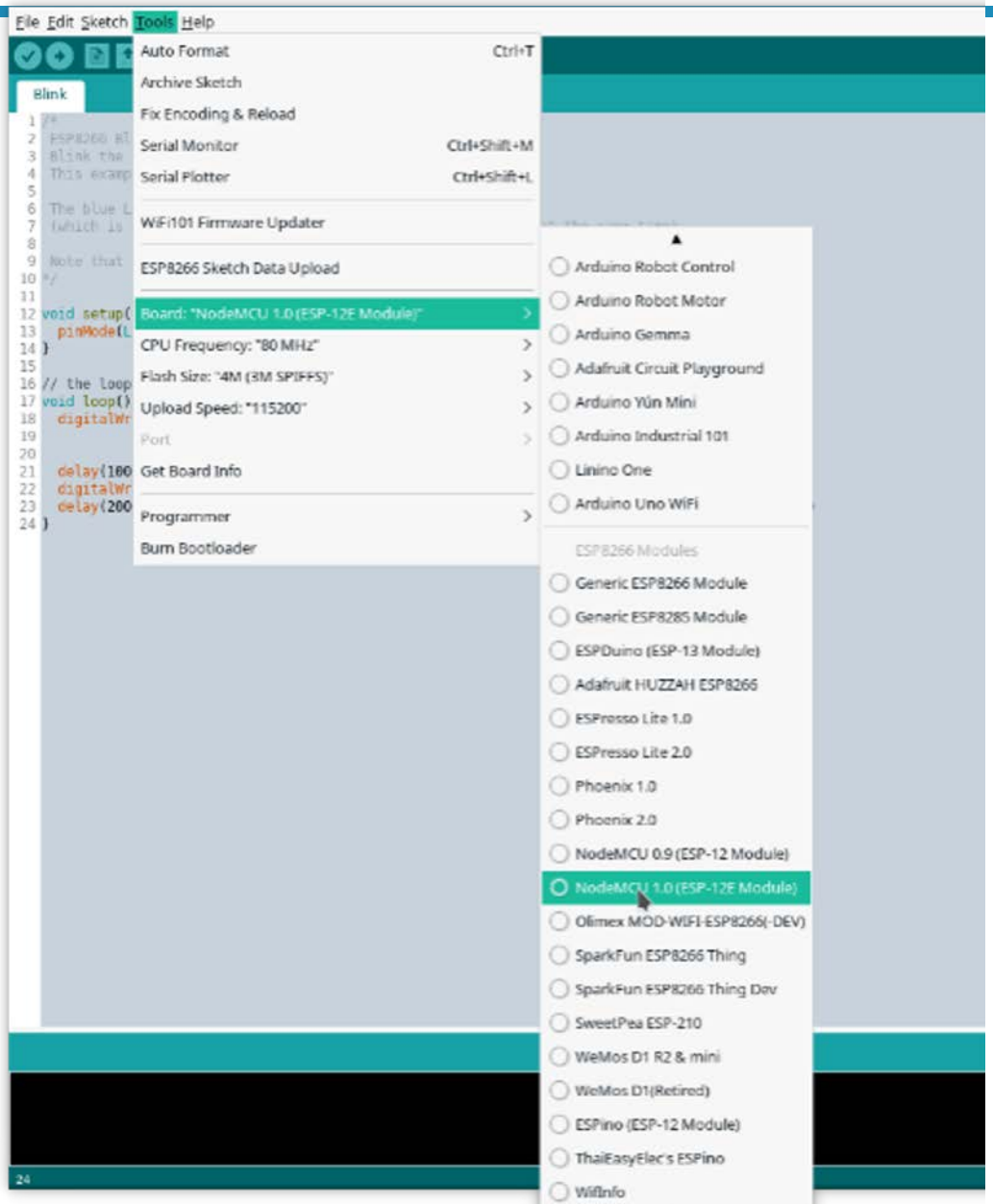
- Search the board manager for *esp8266*
- Install the ESP8266 library





NodeMCU Setup

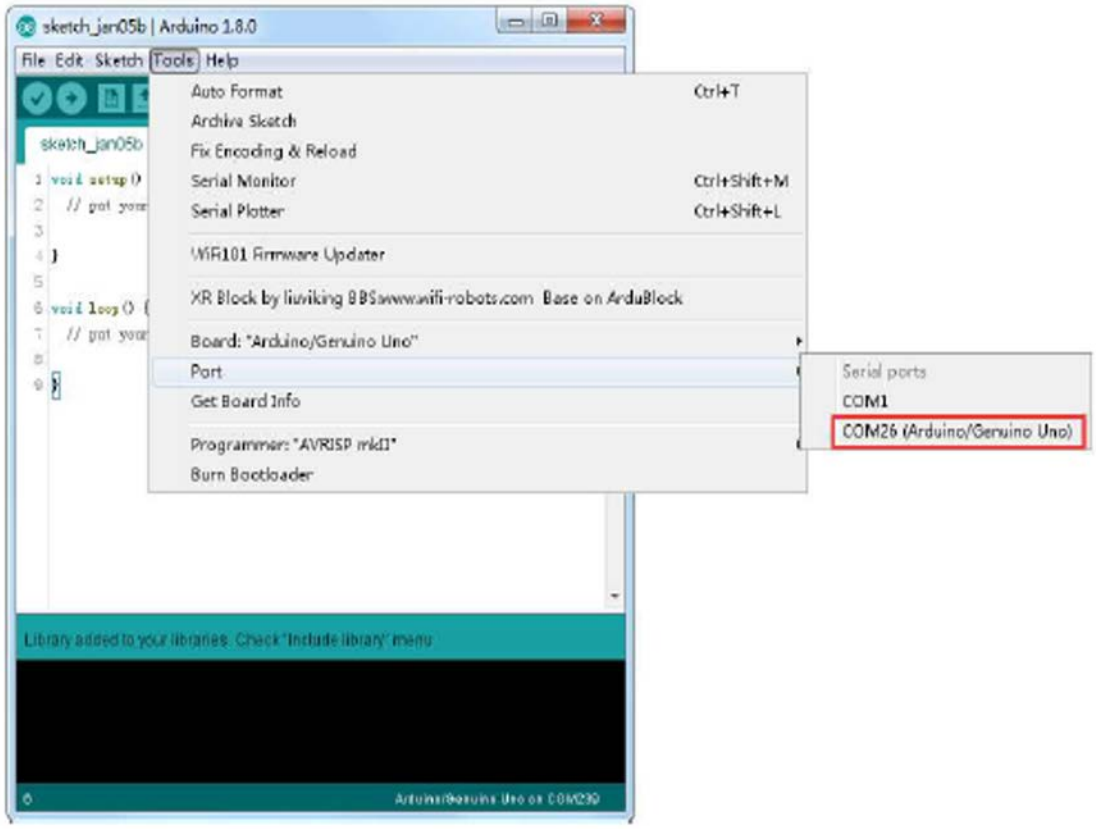
- Select the NodeMCU 2.0 as your board
- Additional settings appear under the board menu
- They can be left as they are
- Higher Upload Speed reduces your upload times

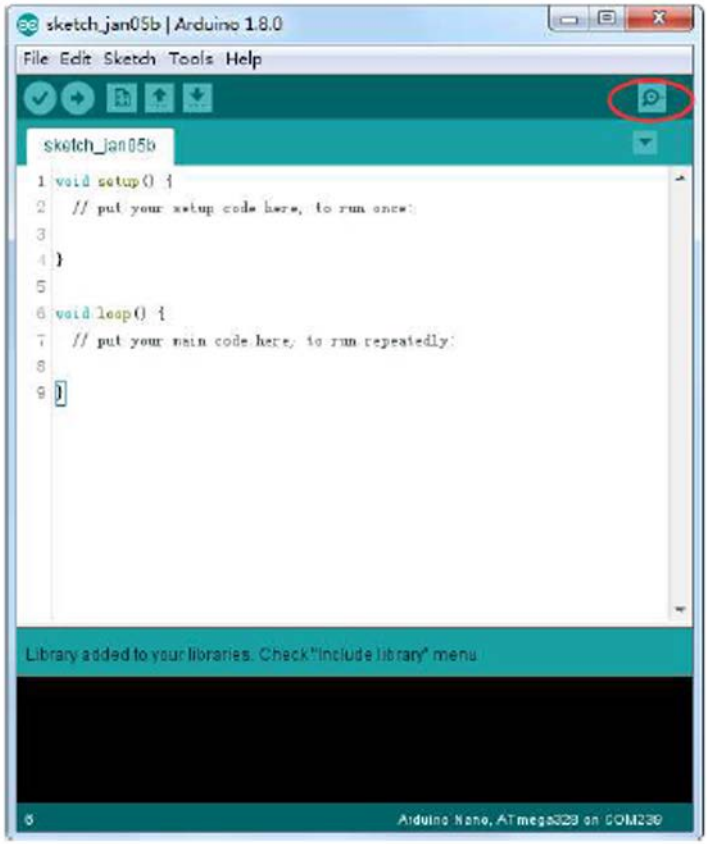


Step 4 - Open Serial Monitor

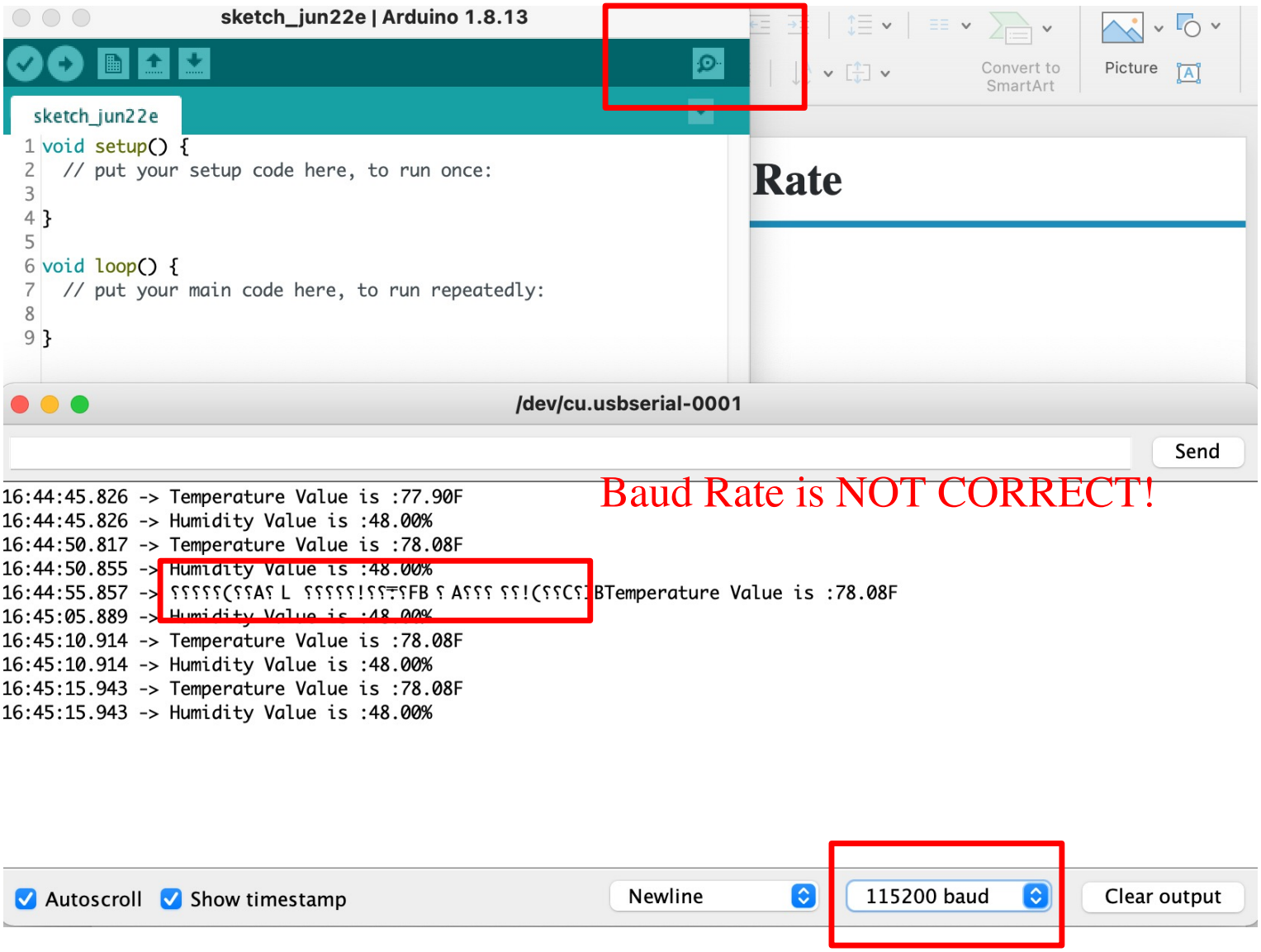
Open Serial Monitor

Arduino Serial Monitor (Windows, Mac, Linux) and Make a Serial Connection
The Arduino Integrated Development Environment (IDE) is the software side of the Arduino platform. And, because using a terminal is such a big part of working with Arduinos and other microcontrollers, they decided to include a serial terminal with the software. Within the Arduino environment, this is called the Serial Monitor.





Check Serial Monitor and Baud Rate



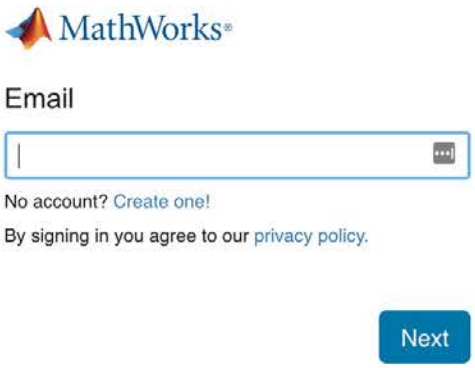
Step 5 - Connect ThingSpeak

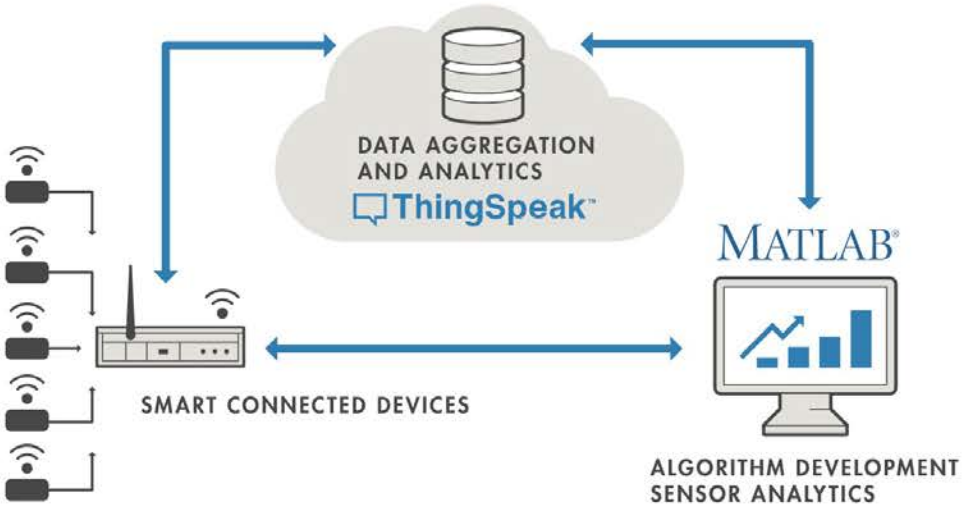
Connect ThingSpeak

https://thingspeak.com/

- Login Page

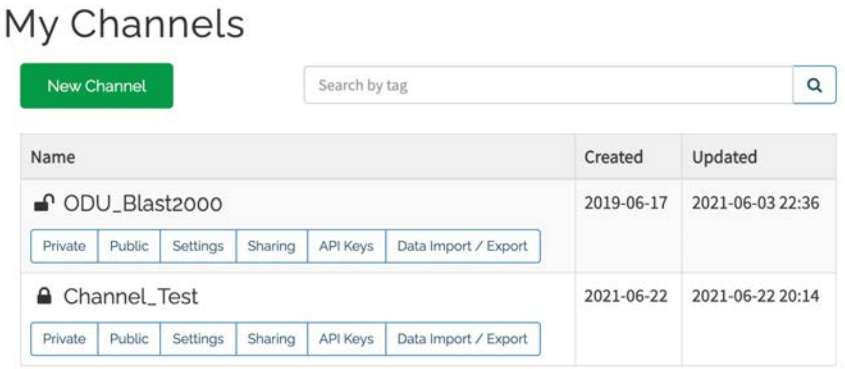
- Email: mkuzlu@hotmail.com
- Password: ODU_Blast2000



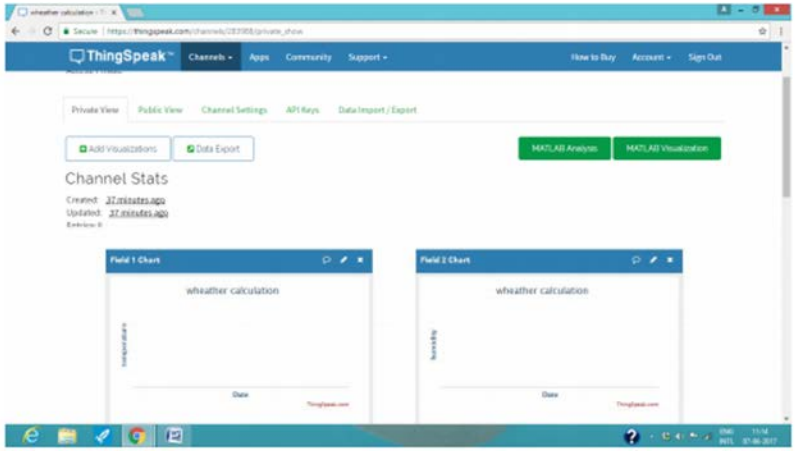


Creating A Channel

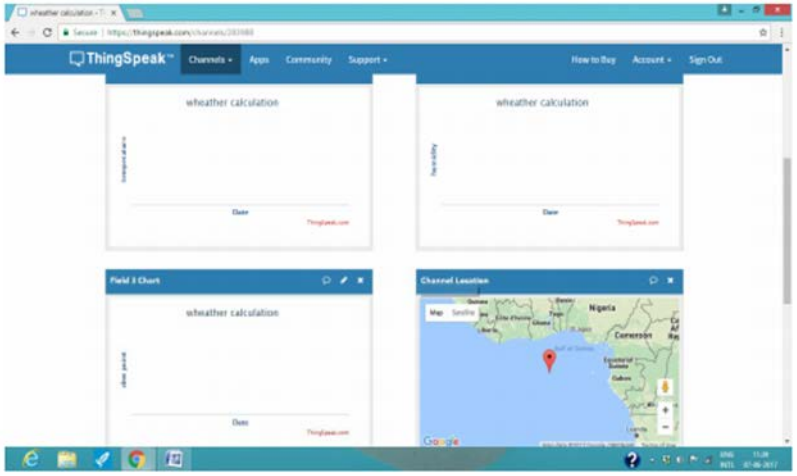
See My Channels



Private View-- It displays the information about your channel that is only visible to you (Recommended)



Public View- if you have chosen to make your channel publicly visible then it will display the selected fields and information



Step 6 - Hands on Activities and Source Code

Hands on Activities and Source Code

Copy&Paste or Download examples, from GitHub:

- Full: https://github.com/muratkuzlu/ODU_BLAST2021

- DHT11 (Recommended): https://github.com/muratkuzlu/ODU_BLAST2021/tree/main/Projects/4_DHT11

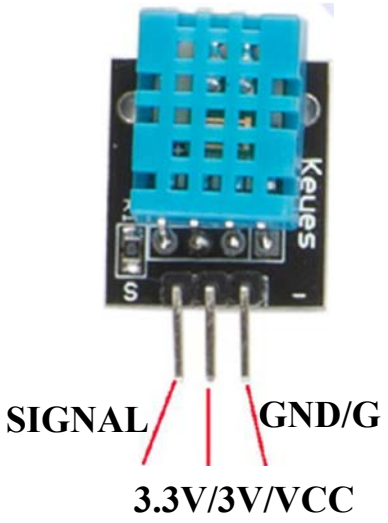
Focus on **end-to-end results, not details**

Start!
Hands-on!

Temperature & Humidity Sensor DHT11

The DHT11 sensor can detect temperature (C and F) & humidity.

The DHT11 detects water vapor by measuring the electrical resistance between two electrodes. The humidity sensing component is a moisture holding substrate with electrodes applied to the surface.
It has everything it requires built into it, so it will work very well with the NodeMCU. This sensor is used in conjunction with the DHT11 Library.



Wiring a switch with Arduino

We will learn how to set up the DHT11 Humidity and Temperature sensor on your NodeMCU. And learn about how the Humidity sensor works, and how to check output readings from the Serial monitor.

Hardware

- NodeMCU
- DHT11 Humidity and Temperature sensor
- Breadboard
- Jumper Wires (Optional)
- Micro USB Cable

Software

- Arduino IDE(version 1.6.4+)
- https://github.com/muratkuzlu/ODU_BLAST2021/tree/main/Projects/4_DHT11

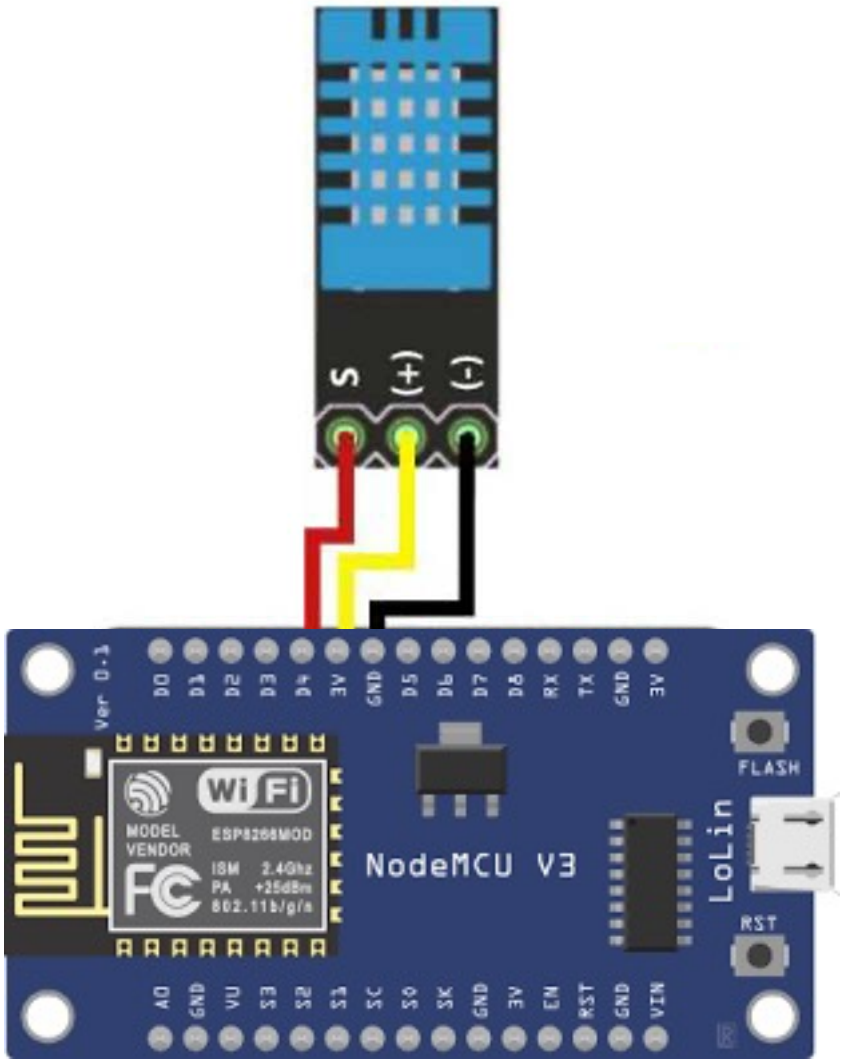
Digital input with Arduino

Set up

Wiring the **DHT11** to the NodeMCU is really easy, but the connections are different depending on which type you have either 3-pins or 4-pins

The **wiring connections** are made as follows:

- **Pin 1 (S or SIG)** of the DHT11 goes into Digital Pin **D4** of the NodeMCU.
- **Pin 2 (VCC or +)** of the DHT11 goes into **+3.3v** of the NodeMCU.
- **Pin 3 (GND or -)** of the DHT11 goes into Ground Pin (**GND**) of the NodeMCU.



Without IoT

Digital input with Arduino

Code

➔ Copy the following code to the IDE

Add the DHTesp.h library

Define Variables

Initialize variables. Runs once

Used to actively control the Arduino board. Run repeatedly

```
#include "DHTesp.h"

int temperature, humidity, k=0, l=0;

#define DHTPIN D4         //pin where the dht11 is connected
DHTesp dht;

void setup()
{
    Serial.begin(115200);
    delay(10);
    //dht.begin();
    dht.setup(DHTPIN, DHTesp::DHT11); // data pin 4
    }

void loop()
{
 static boolean data_state = false;
 float humidity = dht.getHumidity();
 float temperature = dht.getTemperature();
 temperature = CelsiusToFahrenheit(temperature);
 Serial.print("Temperature Value is :");
 Serial.print(temperature);
 Serial.println("F");
 Serial.print("Humidity Value is :");
 Serial.print(humidity);
 Serial.println("%");
 delay(5000);
}
```

Functions

```
//Functions
float FahrenheitToCelsius(float fahrenheit)
{
   float celsius;
   celsius = (fahrenheit - 32.0) * 5.0 / 9.0;
   return celsius;
}


float CelsiusToFahrenheit(float celsius)
{
   float fahrenheit;
   fahrenheit = (celsius * 9.0) / 5.0 + 32;
   return fahrenheit;
}
```

NOTE:
When you check the serial monitor make sure the baud rate and the serial begin number in your code is the same.

➔ Upload

With IoT

Digital input with Arduino and IoT - I

Now we are going to connect to IoT

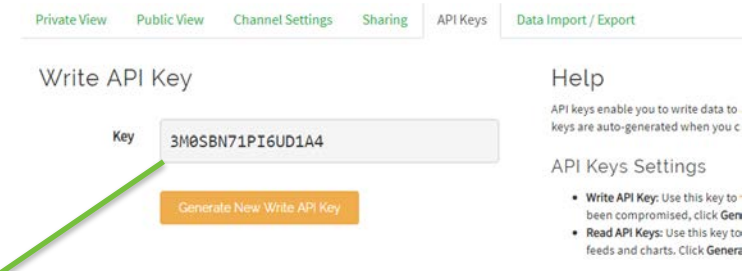
➔ Copy the following code to a new IDE sketch

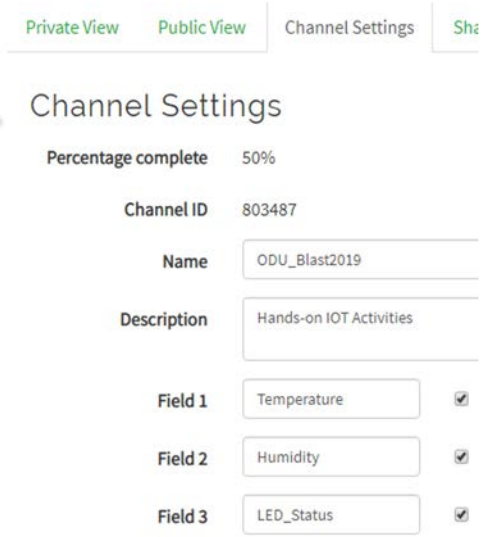
```
#include "DHTesp.h"
#include "ThingSpeak.h"
#include <ESP8266WiFi.h>


const char *ssid =  "AS2L-Room";     // replace with your wifi ssid and wpa2 key
const char *pass =  "as2l214c";
const char* server = "api.thingspeak.com";
const char * myWriteAPIKey = "3M0SBN71PI6UD1A4";  // Enter your Write API key from ThingSpeak
unsigned long myChannelNumber = 803487;
uint8_t temperature, humidity, k=0, l=0;


#define DHTPIN D4          //pin where the dht11 is connected
DHTesp dht;


WiFiClient client;
```





Digital input with Arduino and IoT - II

```
void setup()
{
    Serial.begin(115200);
    delay(10);
    //dht.begin();
    dht.setup(DHTPIN, DHTesp::DHT11); // data pin 4
    Serial.println("Connecting to ");
    Serial.println(ssid);
    WiFi.begin(ssid, pass);
   while (WiFi.status() != WL_CONNECTED)
   {
        delay(500);
        Serial.print(".");
   }
    Serial.println("");
    Serial.println("WiFi connected");


   // Print the IP address
   Serial.println(WiFi.localIP());
   ThingSpeak.begin(client);
}
```

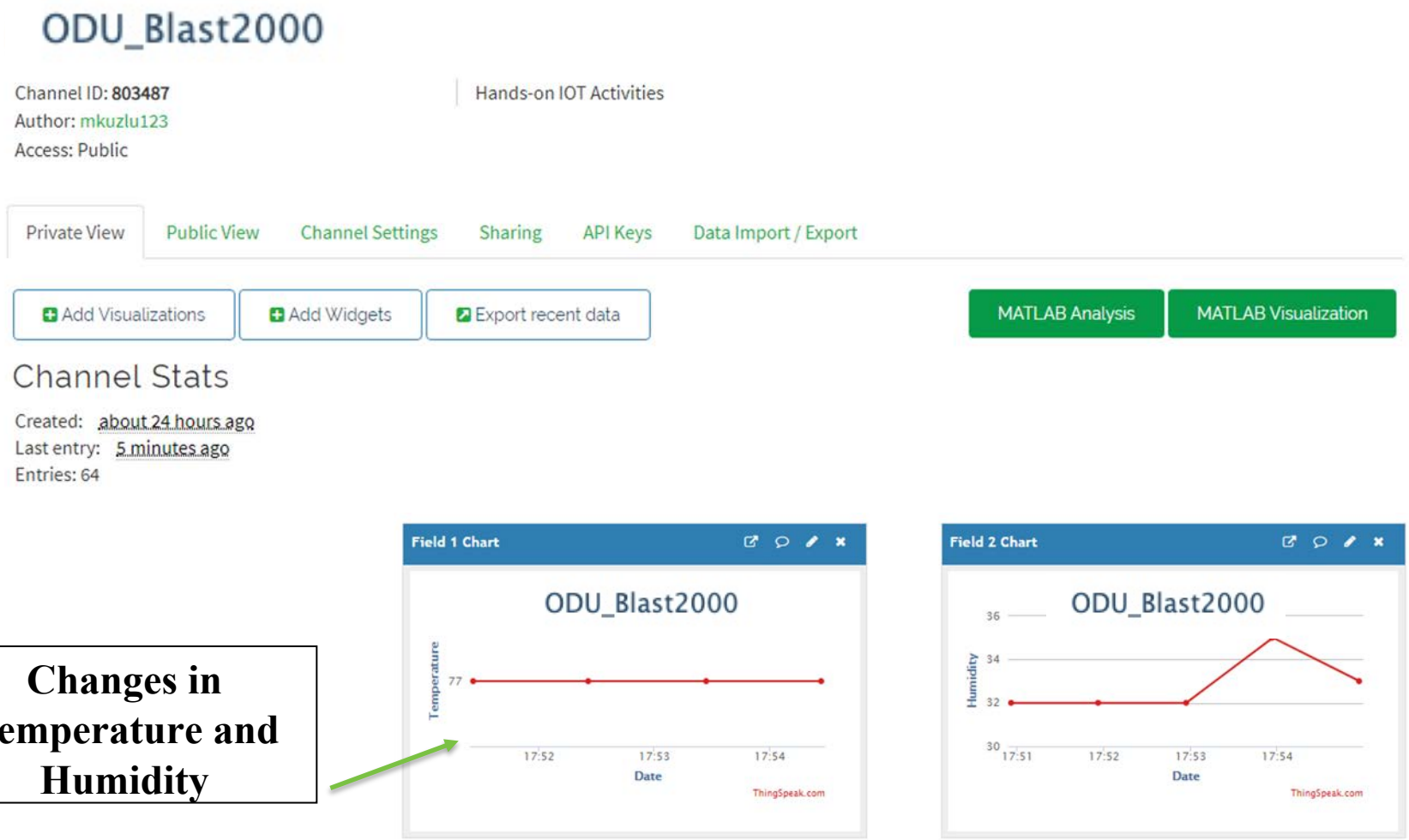
Digital input with Arduino and IoT - III

```
void loop()
{
 static boolean data_state = false;
 float humidity = dht.getHumidity();
 float temperature = dht.getTemperature();
 temperature = CelsiusToFahrenheit(temperature);
 Serial.print("Temperature Value is :");
 Serial.print(temperature);
 Serial.println("F");
 Serial.print("Humidity Value is :");
 Serial.print(humidity);
 Serial.println("%");
 // Write to ThingSpeak. There are up to 8 fields in a channel, allowing you to store up to 8 different pieces of information in a channel.
Here, we write to field 1.
 if(temperature < 255)
 {
    k=temperature;
 }
  if(humidity < 255)
 {
    l=humidity;
 }
 if( data_state )
 {
  ThingSpeak.writeField(myChannelNumber, 1, k, myWriteAPIKey);
  data_state = false;
 }
 else
 {
  ThingSpeak.writeField(myChannelNumber, 2, l, myWriteAPIKey);
  data_state = true;
 }
 delay(30000); // ThingSpeak will only accept updates every 15 seconds.
}
```

➔ Upload

Digital output with Arduino and IoT -IV

➔ Check ThingSpeak



Changes in Temperature and Humidity



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